

Waseem Raza, PhD

Machine Learning Engineer | AI/ML for Wireless & 5G/6G Systems

Wireless AI | 5G/6G | RAN Intelligence | Network Automation

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

AI/ML Engineer for Wireless Networks with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience in 5G/6G systems, network intelligence, propagation modeling, mobility, RAN optimization, anomaly detection, and resilient automation. Apply deep learning, transfer learning, domain adaptation, generative modeling, reinforcement learning, and statistical methods to network-assurance, NR-NTN, CSI compression, digital-twin, outage-management, and self-healing problems. Experienced with Python, PyTorch, TensorFlow, scikit-learn, SQL, large-scale network measurements, 3GPP-compliant simulation, and production-oriented ML validation.

Technical Skills

AI/ML for Wireless: Transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, anomaly detection, classification, and robust model evaluation

Wireless Systems: LTE/4G, 5G-NR, NR-NTN, mobility, RAN, propagation, cell selection, user timing, network measurements, and service assurance

Network Intelligence: Root-cause analysis, outage detection and compensation, self-healing, optimization, resilience, and intelligent automation

Simulation & Digital Twins: 3GPP-compliant simulation, propagation modeling, synthetic data generation, digital twins, large-scale experimentation, and robustness testing

Programming & Data: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, visualization, and statistical analysis

Engineering & MLOps: Git, Docker, CI/CD, reproducible experimentation, APIs, cloud analytics, model validation, and production-oriented workflows

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

02/2025 - Present

Bedminster, NJ

- Lead AI/ML development for E911 and network-assurance diagnostics, applying NLP, domain adaptation, and root-cause classification to operational telecom workflows.
- Develop multimodal diagnostic models using work-order text, location information, network context, and operational signals for VoNR and 5G service assurance.
- Design cross-domain validation and distribution-shift evaluation workflows to improve reliability of ML models across heterogeneous network environments.
- Collaborate with network engineers, software teams, and public-safety stakeholders on AI-enabled automation for critical telecommunications operations.

Advanced-Degrees Intern

AT&T Labs

06/2024 - 08/2024

Austin, TX

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern

Futurewei Technologies

09/2022 - 01/2023

Schaumburg, IL

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern

AT&T Labs

06/2022 - 08/2022

Bedminster, NJ

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant

AI4Networks, University of Oklahoma

09/2019 - 07/2025

Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering

University of Oklahoma, Norman, Oklahoma, USA

09/2019 - 07/2025

MSc, Telecommunication Engineering

University of Engineering and Technology, Taxila, Pakistan

08/2014 - 08/2016

BSc, Telecommunication Engineering

University of Engineering and Technology, Taxila, Pakistan

11/2010 - 06/2014

Selected Projects

SyntheticNET: 3GPP-Compliant Network Simulation

2020-2024

Research Project

- Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks.

Tools & methods: Python, 3GPP, Simulation, Wireless Networks

TurboRAN Experimental Platform

2020-2024

Research Infrastructure

- Multi-band, multi-tier experimental platform supporting evaluation of emerging AI-enabled radio-access-network solutions.

Tools & methods: 5G Testbeds, RAN, Experiment Design, Network Measurement

Certifications

1. **AI for Telecommunications Specialization** — [AI CERTs / Coursera](#) | Completed (3/3) courses | 06/2026 | [Certificate](#)

Focus: Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.

Tools & Skills: AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow

2. **4G Network Fundamentals** — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | 08/2026 | [Certificate](#)

Focus: Foundational study of fourth-generation cellular networks.

Tools & Skills: 4G Networks, Cellular Networks, Telecommunications

3. **Business Considerations for 5G with Edge, IoT, and AI** — [EDUCBA / Coursera](#) | Completed (1/1) courses | 02/2026 | [Certificate](#)

Focus: Explores business considerations for combining 5G connectivity with edge computing, IoT, and AI.

Tools & Skills: 5G, Edge Computing, Internet of Things, Artificial Intelligence, Business Analysis

4. **Machine Learning** — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization

Selected Publications

On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks

2025

Journal article

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin

2024

IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks

2024

IEEE SmartNets

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm)

2022

IEEE Network

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN

2022

IEEE WCNC